

2026 Statistics Graduate Program Comprehensive Exam: Open Book / Analysis Portion

Introduction

Growth charts show selected body measurements in children. Growth charts are used by pediatricians, nurses, and parents to track the growth of infants, children, and adolescents. The x-axis of the chart is age, and the y-axis is the measurement of interest.

This dataset is historical information from a survey on many children. To the best of our knowledge no child is repeated in the dataset at a different age. Data are at <https://grimshawville.byu.edu/ChildGrowth.csv>

Response Variable:

- Weight, in kilograms

Possible Explanatory Variables for the Model:

- Age, in years
- Gender, an integer indicator of gender (1 = male, 2 = female)
- Ethnicity, an integer indicator of 3 groups (unspecified)
- Height, in centimeters

Import:

```
library(tidyverse)
ChildGrowth <- read_csv("https://grimshawville.byu.edu/ChildGrowth.csv")
ChildGrowth <- ChildGrowth %>%
  mutate(Gender = factor(Gender),
         Ethnicity = factor(Ethnicity))
```

Research Questions/Analysis Goals

Interpretation:

- What are the characteristics of the growth chart (effect of Age)?
- How many growth charts should we provide to doctor's offices? Do we need different charts for each gender and/or ethnicity?

Prediction:

- Compare in-sample and out-of-sample prediction performance.
- Are there any biases in prediction? That is, are there any children that are generally over- or under-estimated?

Analysis

You will fit two models and your code and report will include the details on these two models. One analysis is prescriptive while the other will be analyzed reflecting your decisions about the analysis objectives.

Model A:

Fit a regression model for $\log(\text{Weight})$ that satisfies the following:

- Growth chart (effect of Age) is nonlinear and monotone increasing
- Write the model as a “cell means model” with a different growth curve for each gender and ethnicity
- Note: Ignore Height as an explanatory variable in Model A

Use the model to perform the following analysis (conclusions use 0.01 significance level or 99% confidence level):

- Test H_0 : No gender AND no ethnicity effect
(full points require using a full and reduced model ANOVA test)
- Test H_0 : Parallel growth curves for all gender and all ethnicity
(full points require using a C vector or matrix)
- Test H_0 : No difference between female and male growth curves (holding ethnicity constant)
- Test H_0 : No difference between ethnicity growth curves (holding gender constant)
- Test H_0 : Growth curves are linear
- Make a growth chart of Age vs. $\log(\text{Weight})$ for an average child with Gender = 1 and Ethnicity = 3. Include uncertainty on the growth chart.
- Estimate the difference between female and male growth curves for Ethnicity = 2 at Age = 10. Untransform the estimate, and provide an interpretation that includes uncertainty.

Model B:

Accurate predictions are crucial. (Imagine a parent who is worried if their child's weight is normal). A general sense of important factors and comparison to other groups is helpful in believing the prediction.

You may choose any model, but you will support your modeling choice and explain the advantages and disadvantages of the model you selected in terms of how well it addresses the research questions. There may be multiple ways to adequately go about answering the question.

Just to be clear, Height may be used as an explanatory variable in Model B.

Report Instructions

Report Page Limit: 8-10 pages (Sections Model A and Model B should be 5-6 pages that focus on important model tuning practice / results, reporting prediction performance, and practically significant effect results / interpretations)

Report Sections & Contents:

- **Introduction:**
 - Writing Prompts:
 - What are growth curves?
 - How are they used by parents? doctors? researchers in early childhood development?
 - What are the properties of growth curves?
 - What factors might result in different growth curves?
- **Data:**
 - Writing Prompts:
 - Table 1: Summary Statistics with discussion
 - Data features (most important features learned from EDA) that impact Model A estimation and inference and decisions about the Model B analysis choice (discussion with some graphics)
- **Model A:**
 - Write the “cell means model” explaining all notation
 - Explain why you chose your approach to modeling nonlinear growth curves. Share other ways this could be achieved and point out why your approach is better.
 - Describe how you tuned the model.
 - Report each of the hypothesis tests. Write the hypothesis in the model parameters, share the reduced model or C vector or matrix used, report the test statistic and p-value, and provide an interpretation for the target audience of health care providers, child development researchers, and public health professionals wanting to understand growth curves and make positive changes with good public policy.
 - Share the growth chart of Age vs. $\log(\text{Weight})$ for an average child with Gender = 1 and Ethnicity = 3. Include uncertainty on the growth chart.
 - Report the estimated difference between female and male growth curves for Ethnicity = 2 at Age = 10. Provide an interpretation for the target audience that includes uncertainty.
- **Model B:**
 - Explain why you chose this model over other possibilities. Specifically address how it achieves the analysis goals.
 - Describe the model structure and key assumptions.
 - List the predictors used and any transformations or feature engineering.
 - Describe how you tuned the model.

- Perform 10-fold cross-validation and report out-of-sample RMSE of Weight in a table.
- Share the variable importance of the final form of the model after tuning / variable selection / etc. and comment on how the most important variable may be impacting predictions.
- Use the final form of the model after tuning / variable selection / etc. to describe the difference in male and female growth, in general. Also, Model A asked for the estimated difference between females and males for Ethnicity = 2 at Age = 10, with uncertainty. Can your method do this too?
- Also share at least one aspect of the model that would be of interest to a non-statistician. This could be an interesting model feature or an effect of practical interest. (Examples: Is the growth curve nonlinear? Are growth curves parallel by ethnicity?)
- **Compare Models and Choose One**
 - Writing Prompts:
 - Discuss how each model estimates the growth rate effect (that is, the Age effect). Specifically how smooth / wiggly and how monotone is the estimated Age effect.
 - Discuss how each model estimates differences in Gender and Ethnicity.
 - If you had to choose just Model A or Model B to publish / share for healthcare providers following evidence-based practice, which would it be? Your choice should use the interpretability / predictability trade-off and be written for a statistician / data scientist comfortable with both model strengths & weaknesses.
 - Note: Full points require "Excellent Data Viz."
- **What Would Improve The Model?**
 - Writing Prompts:
 - If there are outliers / influential observations, identify them and address (speculate) on their impact on Model A and/or Model B.
 - Share "data you wish you had" (rows & columns) and comment on how that data would be used to improve Model A and/or Model B.
 - Share features of the variance that may deviate from the independence assumption.

Grading Note: While code will be turned in with the report, assume graders will typically look at the code when something important is poorly explained in the report or a critical technical detail is omitted from the report.

AI policy: During the exam period, you may not discuss the exam with anyone. The exam is open notes, open internet, and AI tools are allowed. You are encouraged to use generative AI to support your work efficiently. However, the exam is meant to demonstrate your own judgment, reasoning, and statistical practice. You are responsible for everything you submit and must ensure that any AI-assisted content is correct and high quality.